

Question 1:

Violated ACM Code of Ethics:

- **1.2** – Avoid harm
- **1.3** – Be honest and trustworthy
- **2.5** – Give comprehensive and thorough evaluations of computer systems and their impacts
- **2.6** – Perform work only in areas of competence
- **3.1** – Ensure that the public good is the central concern during all professional computing work
- **3.2** – Encourage acceptance of and evaluate fulfillment of social responsibilities by members of the organization or group

Question 2:

To ethically balance the AI model's prediction with the radiologist's assessment:

1. **Informed Consent:** Ensure the patient understands the AI model's role and limitations.
2. **Collaboration:** The clinical team should discuss the case together, weighing both the AI model's prediction and the radiologist's judgment.
3. **AI as Support:** Use the AI as a tool, not a sole decision-maker, acknowledging its lack of clinical context.
4. **Trust in Expertise:** Prioritize human expertise over AI when reasoning is opaque.
5. **Patient Autonomy:** Involve the patient in the decision, outlining risks of both AI predictions and biopsy.
6. **Second Opinion:** Consider getting a second opinion for more clarity.
7. **Improve AI Transparency:** Develop explainable AI (XAI) for future decisions.

This approach ensures a balanced, patient-centered decision-making process.

Question 3:

Two major ethical issues are:

1. Unauthorized use and misrepresentation of Nayeem's video.
2. Violation of informed consent, exposing Nayeem and villagers to harm.

Although consent was given for academic use, the video was edited and shared publicly with misleading political content.

Nayeem should:

1. Report the misuse to the social media platform and request removal.
2. Inform his university and the villagers about the situation.
3. Seek legal or institutional support if needed to protect everyone's rights.

Question 4:

No, it is not ethical for the app to collect and share sensitive data (like GPS and heart rate) based on vague consent hidden in lengthy terms. User consent must be informed, explicit, and easy to understand. Users often skip reading lengthy agreements, which makes such data collection non-transparent and potentially exploitative.

Ethical issues to consider:

Informed Consent: Inform users what data is collected and how it is used.

Data Minimization: Only collect what's necessary.

Transparency: Provide a simple summary of data usage.

Right to withdraw: Let users control third-party sharing.

Data Security: Protect user data from unauthorized access.

Sample consent letter:

Consent for Fitness App Data Collection

Dear User,

To provide personalized fitness services, we collect:

- GPS and location data
- Heart rate and activity patterns

You have the right to:

- Opt out of data sharing
- Request the deletion of your data
- Review what data we store about you

☐ I consent to the use and sharing of my data as described above.

☐ I do not consent and prefer limited app features.

Name: _____ Date: _____ Signature: _____

Question 5:

No, pursuing environmental goals through harmful methods is not acceptable. Technologies that pollute, deplete resources, or displace communities defeat the purpose of sustainability. True environmental responsibility requires minimizing waste, avoiding toxic byproducts, and using ethical, clean sources. Instead of shifting the burden elsewhere, solutions should focus on recycling, local eco-friendly materials, and community involvement to ensure long-term, ethical sustainability.

Full environmental impact assessment – Ensure no toxic leakage or underground damage occurs

Sourcing eco-materials ethically – Avoid importing from factories with known pollution issues

Engage the local community – Prevent displacement and include residents in planning

Adopt closed-loop recycling systems – Especially for infrastructure and electronics

Focus on transparency and accountability – Don't use technology to manipulate or surveil without consent

